Alamein University
Faculty of Computer science & Engineering

Computer Vision
Course Code: CSE383

Final Project: Computer vision course:

Smart Classroom Assistant

Project Description

This project focuses on building a **Computer Vision-based Smart Classroom Assistant** that monitors students, analyzes their behavior, and enhances interaction using Speech Recognition and NLP.

The system primarily relies on **Computer Vision** to understand the classroom, while Speech and NLP modules are integrated to enrich the interaction.

. The assistant can take attendance automatically, analyse students' moods, convert spoken questions into text, and classify them into specific topics.

Project Modules:

Computer Vision Module

• Use face recognition (OpenCV + face_recognition library) to identify

students and take attendance automatically.

• Add emotion detection (using pre-trained CNN or fer library) to

estimate students' impressions (e.g., happy, tired, sad).

• Output: Student name, presence status, and mood impression.

Speech Module

• Use a microphone to capture student speech.

• Apply speech recognition (Google API) to convert spoken questions into text.

- Example: Voice input 'Can you explain convolution?'
- Output text: 'Can you explain convolution?'

NLP Module

- Take the transcribed question and classify it into a topic (e.g., Math, Data Science, Computer Vision).
- Use TF-IDF with Scikit-learn or pre-trained NLP models from Hugging Face.
- Output: The predicted topic for each student question.

Integration Workflow

- At the start of the class: Computer Vision takes attendance and detects student moods.
- When a student asks a question: Speech Module converts the spoken audio into text.
- NLP Module classifies the question into a subject category.
- All outputs are stored together in a structured log (JSON/CSV format).

Example Output

{ "student": "Ahmed Ali", "attendance": "Present", "mood": "Tired", "question": "Can you explain convolution?", "topic": "Computer Vision" }

Suggested Tools

- Computer Vision: OpenCV, face_recognition, fer
- Speech: SpeechRecognition library
- NLP: Scikit-learn (TF-IDF + classifier)

Conclusion

By the end of this project, students will have created an integrated Smart Classroom Assistant capable of marking attendance, recording moods, transcribing spoken questions, and classifying them into topics. This project combines practical skills in Computer Vision, Speech Recognition, and NLP, preparing students for real-world AI applications

Important Details

- **Team Formation:** Each team must consist of **6 students** (min 4 students)
- **Deadline:** Sunday, **May 9, 2026**. (Canvas)
- **Discussions Date:** Monday, **May 10, 2026** (*during the sections time)*.

Submission Requirements:

- All **project code files** must be submitted.
- A **presentation** explaining the project is required.
- **All team members** must participate in submitting the required files.

Best of luck to everyone!